

BI:TASK-01:Exploratory Data Analysis (EDA) using Zara Sales Performance Analysis Dataset

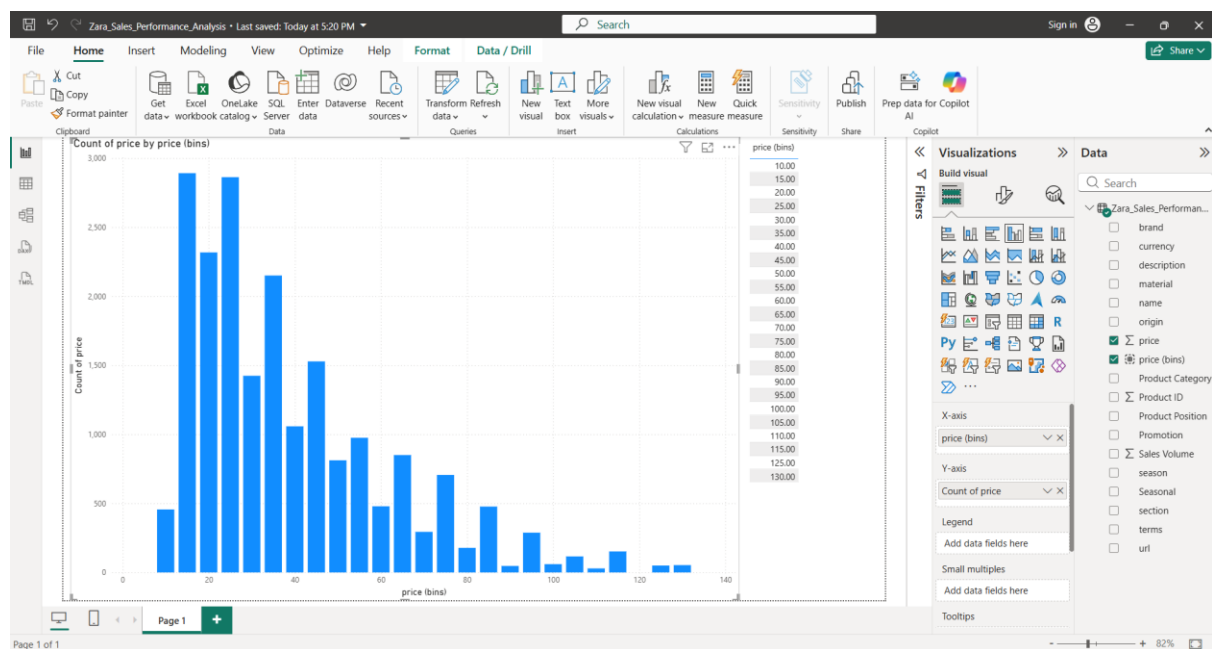
Submitted by : Neha Rose Biju

First we have to load the dataset which contain the sales data of zara

The screenshot shows the Power BI Desktop interface with the 'Zara_Sales_Performance_Analysis.csv' data source being loaded. The data source selection dialog is open, showing the file origin, delimiter (Semicolon), and data type detection (Based on first 200 rows). The data preview shows columns: Product ID, Product Position, Promotion, Product Category, Seasonal, Sales Volume, brand, and url. The Visualizations pane on the right shows the 'Build visual' button and a list of available visualizations.

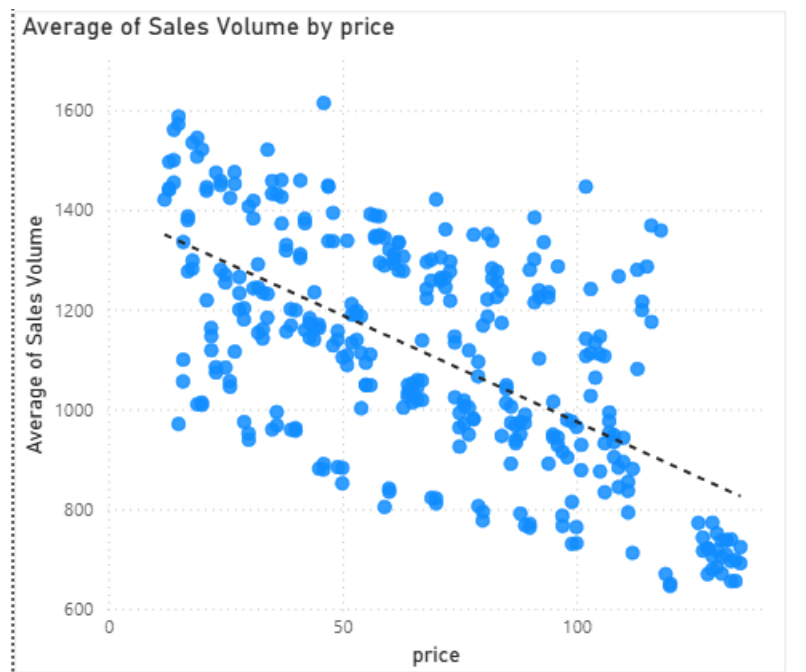
Product ID	Product Position	Promotion	Product Category	Seasonal	Sales Volume	brand	url
185102	Aisle	Yes	clothing	Yes	1243	Zara	https://www.zara.com/us/en/basic-puffe
188771	Aisle	Yes	clothing	No	1429	Zara	https://www.zara.com/us/en/taxedo-jack
180179	End-cap	Yes	clothing	Yes	1168	Zara	https://www.zara.com/us/en/slim-fit-suit
112917	Aisle	Yes	clothing	No	1348	Zara	https://www.zara.com/us/en/stretch-suit
192936	End-cap	Yes	clothing	Yes	1602	Zara	https://www.zara.com/us/en/double-face
117590	End-cap	Yes	clothing	Yes	1282	Zara	https://www.zara.com/us/en/contrasting
189118	Front of Store	No	clothing	No	688	Zara	https://www.zara.com/us/en/faux-leathe
182157	Aisle	Yes	clothing	Yes	1711	Zara	https://www.zara.com/us/en/suit-jacket
141861	Aisle	No	clothing	Yes	857	Zara	https://www.zara.com/us/en/100-wool-s
137121	Aisle	No	clothing	No	769	Zara	https://www.zara.com/us/en/100-feathe
113143	Aisle	Yes	clothing	Yes	1696	Zara	https://www.zara.com/us/en/herringbon
140028	Aisle	Yes	clothing	No	1538	Zara	https://www.zara.com/us/en/oversized-c
134693	Aisle	No	clothing	Yes	822	Zara	https://www.zara.com/us/en/leather-bik
151396	Front of Store	No	clothing	No	840	Zara	https://www.zara.com/us/en/cropped-le
132889	Aisle	Yes	clothing	Yes	1569	Zara	https://www.zara.com/us/en/faux-leathe
152174	End-cap	No	clothing	Yes	952	Zara	https://www.zara.com/us/en/faux-leathe
129906	Aisle	No	clothing	Yes	915	Zara	https://www.zara.com/us/en/faux-suede
195879	Front of Store	Yes	clothing	Yes	1523	Zara	https://www.zara.com/us/en/denim-burr
155050	Aisle	Yes	clothing	Yes	1305	Zara	https://www.zara.com/us/en/boucle-text
194410	End-cap	No	clothing	No	840	Zara	https://www.zara.com/us/en/suit-jacket

(a) What is the price range that concentrates most of Zara's products?



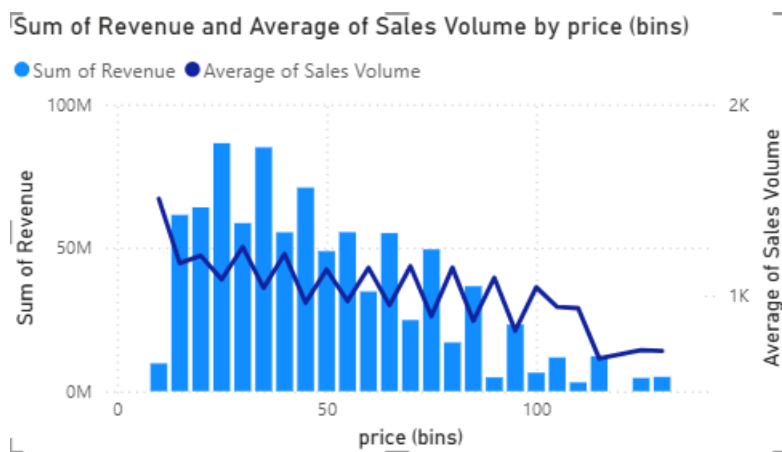
The histogram shows that Zara's products are predominantly concentrated in the lower price ranges. The highest number of products falls between ₹15-₹20 and ₹25-₹30, each containing nearly 2891 and 2862 products respectively.

(b) Do higher-priced products sell proportionally fewer units?



The scatter plot shows that products with lower prices usually have higher sales volumes, while products with higher prices generally have lower sales volumes. This indicates that higher-priced products tend to sell fewer units.

(c) Is there an “ideal” price range that balances sales volume and total revenue?



The chart suggests that the ₹20–₹35 price range provides the best balance between sales volume and total revenue. Products in this range generate the highest revenue while maintaining relatively high average sales volume. Lower-priced products achieve good sales volume but contribute less revenue per product, whereas higher-priced products experience a decline in both revenue and average sales volume.

(d) Does the average price differ significantly between sections (MAN, WOMAN)?

The average prices of products in the MAN and WOMAN sections are nearly identical. Although the MAN(42.13) section has a slightly higher average price than women(41.86), the difference is very small. Therefore, there is no significant difference in the average product prices between the two sections.

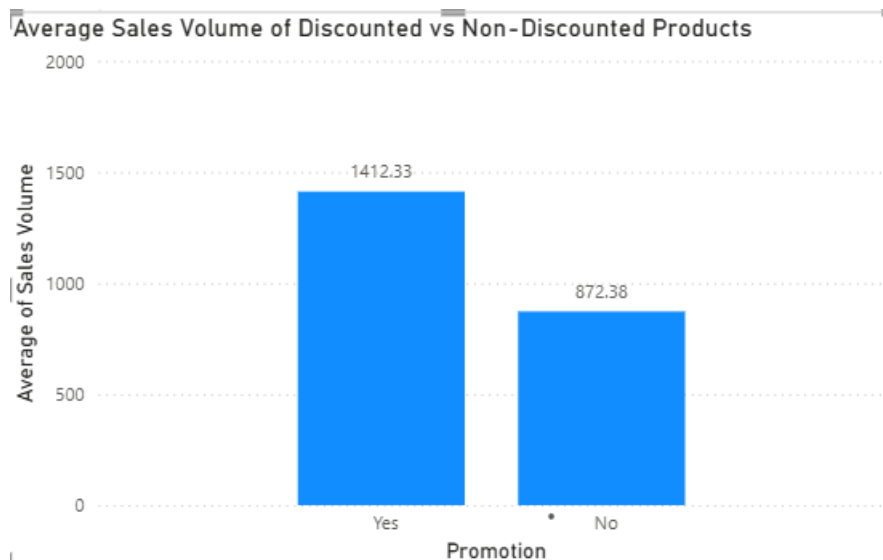


(e) Do discounted products significantly reduce their average price?



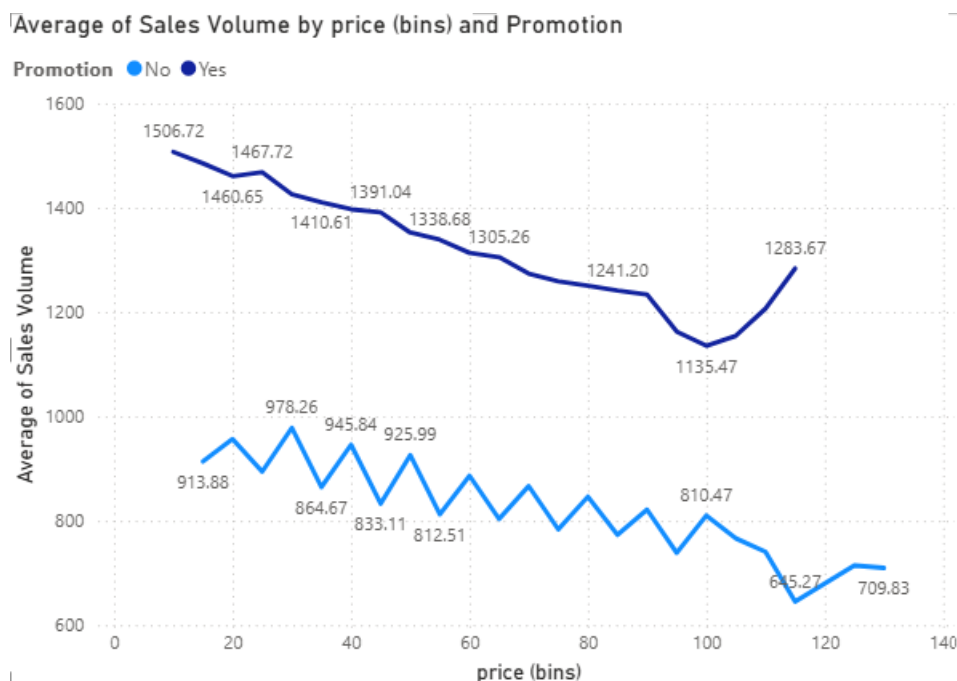
Discounted products have a lower average price than non-discounted products. The average price decreases from 45 to 38, indicating that promotions effectively reduce the average selling price. Therefore, discounted products are generally sold at lower prices than non-discounted products.

(f) Do discounted products have, on average, higher sales volumes?



The chart indicates that discounted products achieve a significantly higher average sales volume than non-discounted products. Products on promotion sell an average of 1412.33 units, while products without promotion sell 872.38 units on average. This suggests that discounts have a positive impact on customer demand and increase sales volume.

(g) Is the difference in sales volume between discounted and non-discounted products greater within a specific price range?

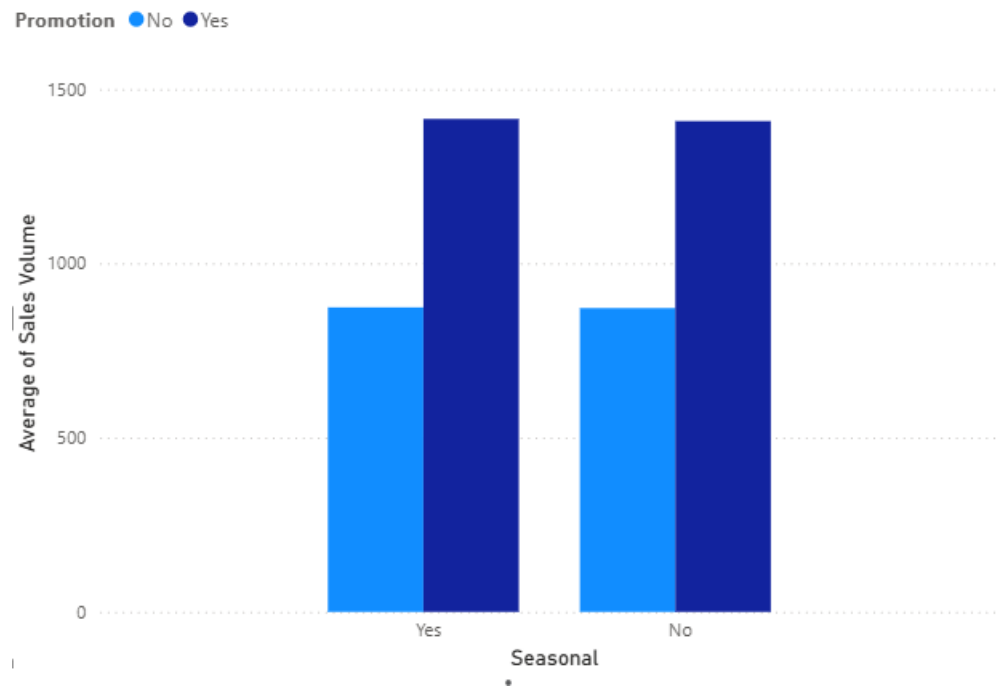


The line chart shows that discounted products consistently have higher average sales volumes than non-discounted products across almost all price ranges. The difference is greatest in the lower and mid-price ranges (approximately ₹15–₹50), where promotional products sell significantly more units. As the price increases, the gap becomes slightly

smaller, but discounted products continue to outperform non-discounted products in terms of average sales volume. This suggests that promotions are most effective for products in the lower and mid-price ranges.

(h) Do promotions affect seasonal products more than non-seasonal ones?

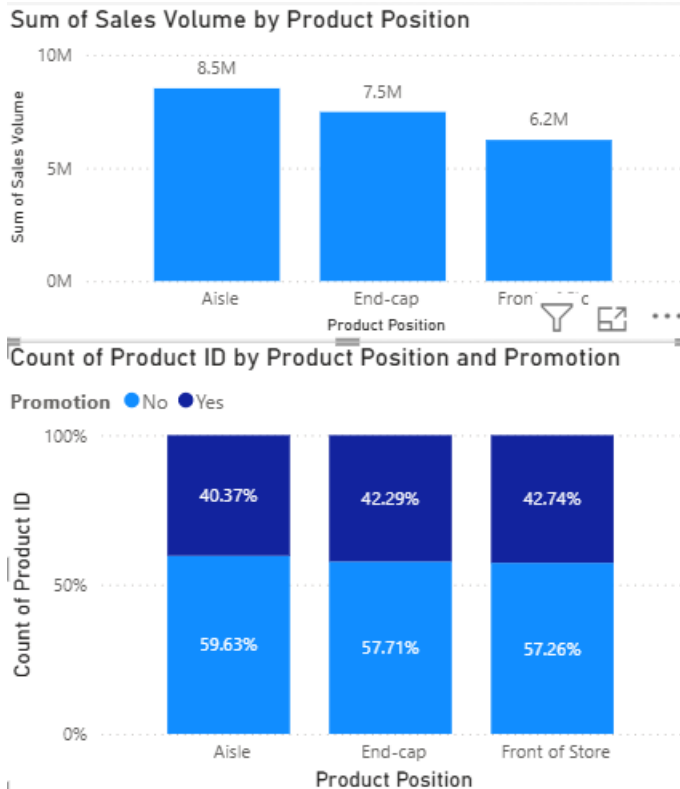
Average of Sales Volume by Seasonal and Promotion



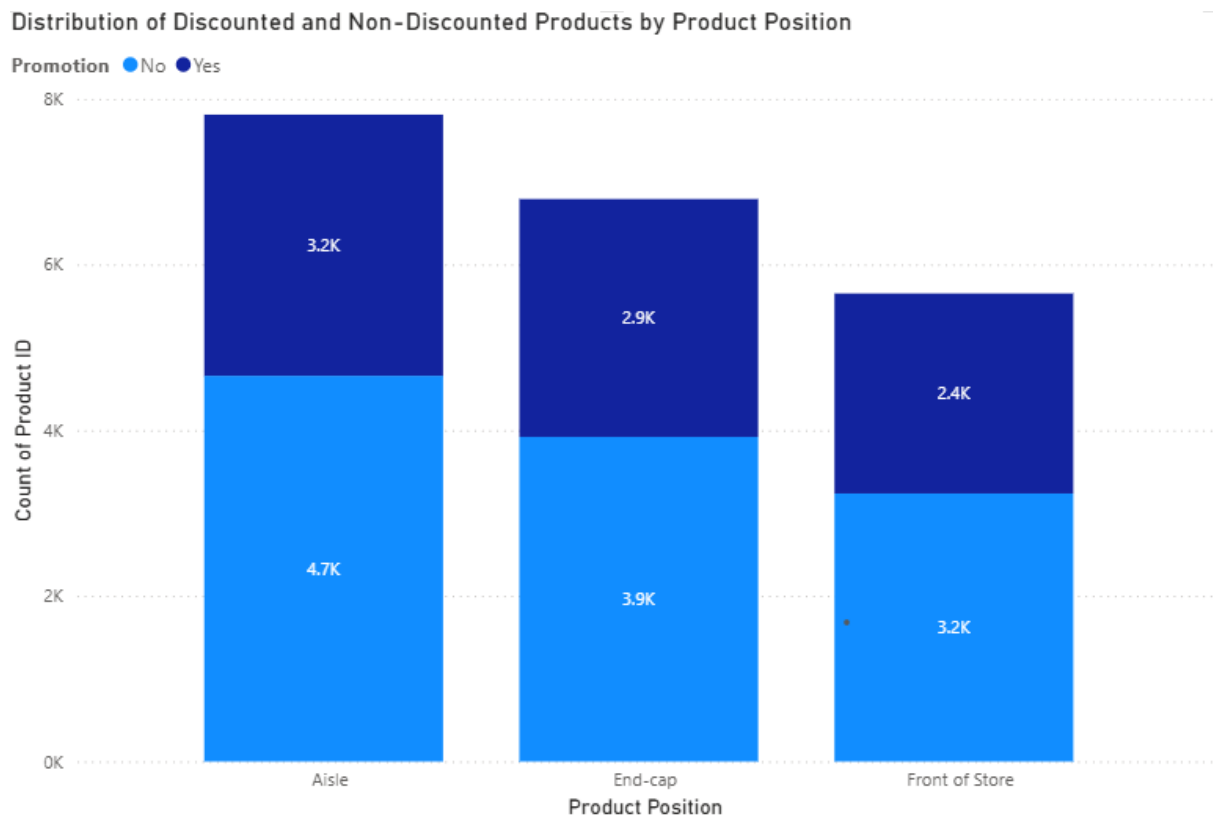
Promotions increase the average sales volume for both seasonal and non-seasonal products. However, the difference between promoted and non-promoted products is almost the same in both groups. Therefore, promotions do **not** appear to have a stronger effect on seasonal products than on non-seasonal products. They improve sales volume by a similar amount regardless of seasonality.

(i) Is there any relationship between the percentage of discounted products and total sales?

The comparison between the percentage of discounted products and total sales does not show a strong positive relationship. Although the Front of Store has the highest percentage of discounted products (42.74%), it records the lowest total sales volume (6.2M). In contrast, the Aisle has the lowest percentage of discounted products (40.37%) but achieves the highest total sales (8.5M). Therefore, the results suggest that a higher percentage of discounted products does not necessarily lead to higher total sales, and other factors such as product placement or customer traffic may also influence sales performance.



(j) Do discounted products appear more frequently in a specific Product Position (e.g., End-cap or Front of Store)?



Discounted products are found in all product positions, but they appear most frequently in the Aisle, followed by the End-cap, and least frequently at the Front of Store. This suggests that the Aisle is the most common location for discounted products. However, non-discounted products are also more numerous in every product position, indicating that promotions are distributed across all locations rather than being concentrated in a single display

Final Dashboard

